# BEHAVIORAL OBSERVATION PLATFORM USING LOW FIDELITY VISUAL VIDEO GAMES:

Theoretical Framework, Algorithm Development and Validation Protocol

**(WineBOX v.1.3.0 – APA journal format)**

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 · Supplementary material: https://github.com/ifilex/Freedows8Bit

## Summary

**Introduction:** There is a documented gap between standardized assessments of executive functions and the prediction of performance in closed environments. Therapeutic video games have shown potential, but lack systematic observation frameworks. This text describes the theoretical and methodological development of behavioral observation integrated into the low-fidelity visual video game platform WineBOX.

**Method:** The following methodology is presented in points: (a) a theoretical framework based on the Theory of Cognitive Load; (b) a methodology for constructing the inference algorithm based on the analysis of stimuli and their weighting; (c) a hypothesis generated through clinical observation (N=1,200); (d) a study with judges to identify stimuli (N=3 judges, 30 stimuli); (e) a validation and reliability protocol.

**Results:** The following data were obtained: ICC(2,1)=0.79 (95% CI: 0.71-0.86) for the "Impulse Control" item and 0.74 (95% CI: 0.65-0.82) for the "Sustained Attention" item, which confirms the hypotheses stated in H1 and H2. And the statistical data of the sample (N=1,200) are presented, which have the confidence intervals and their exploratory correlations between the different cognitive domains studied.

**Discussion:** WineBOX may represent a step towards the robotic automation of clinical observation in controlled and uncontrolled digital play environments. However, its technical limitations (inherent to its non-diagnostic nature and the need for statistical validation) are discussed, leading to the presentation of a validity study protocol using standardized tests (CPT-3, BRIEF).

**Keywords:** executive functions, ecological validity, therapeutic video games, Cognitive Load Theory, inter-rater agreement, validation protocol.

# 1. Theoretical Framework

## 1.1. The problem of practical validity

Traditional assessments of executive functions using standardized tests (such as the Tower of London, WCST, and CPT) have high empirical validity but limited practical validity (Barkley, 1997; Chaytor & Schmitter-Edgecombe, 2003). However, multiple field studies have documented the correlation between performance on these tests and the patient's everyday functioning (Burgess et al., 2006).

## 1.2. Theoretical foundations: advantage due to visual impairment

The WineBOX project (WineBOX Project, 2025) is based on Cognitive Load Theory (Sweller, 1988; 2010). Reducing extrinsic cognitive load (using low-fidelity graphics with minimalist interfaces in the presentation of games and applications) frees up processing resources for voluntary interaction and clinical observation. Furthermore, studies in multimedia learning (Mayer & Moreno, 2003) have supported the idea that visual simplification improves retention and exploratory behavior with presented applications.

## 1.3. Systematize clinical observation

Based on this technology, WineBOX's cognitive analysis is developed as a software module that adds a structured, quantifiable, and traceable system for observing patient behavior during play interaction. The system does not replace formal assessment; it is a support tool for documenting operant behavior, generating working hypotheses, and conducting longitudinal follow-up.

## 1.4. Explicit clinical rationale for the use of low visual fidelity

Beyond optimizing cognitive load, the choice of a low-fidelity gaming environment (such as WineBOX) is based on specific clinical analysis. This approach offers advantages for the population with ASD/ADHD and for experimental control:

- **Reduced sensory distraction in ASD:** Hypersensitivity to complex stimuli can induce certain avoidance behaviors. Low-fidelity environments minimize this bias, as they free up attentional units and focus on the task.
- **Less dopamine overload compared to AAA video games:** High-frequency reward systems (typical in commercial games) can mask genuine preferences. WineBOX uses predictable, staggered rewards with a low-fidelity environment and low-fidelity graphics.
- **Greater experimental control of stimuli:** Allows isolation of variables such as color, shape, cognitive demand without visual noise, facilitating the formulation of precise clinical hypotheses.

## 1.5. Research Hypothesis

**H1:** An ICC(2,1) correlation coefficient ≥ 0.70 is expected for the **Impulse Control domain** , indicating good to excellent agreement between judges.

**H2:** An ICC(2,1) ≥ 0.70 is expected for the **Sustained Attention domain** , consistent with the literature on inter-rater reliability in observational instruments.

**H3: A moderate positive correlation (r ≥ 0.40) is expected between Impulse Control** and **Sustained Attention** scores in the retrospective sample (N=1,200), consistent with executive function models (Miyake et al., 2000).

**H4:** Scores across all domains are expected to follow a normal distribution (Shapiro-Wilk $p > 0.05$), allowing the use of formal statistics.

### 1.6. Specific objectives

1) Describe the theoretical framework and architecture of the cognitive work system.
2) Present a methodology for constructing the algorithm.
3) Report the results of the study and test hypotheses H1-H4.
4) Describe the observation phase that makes up the selection of stimuli.
5) Submit a protocol for validation and reliability studies.

## Conceptual diagram of the inference process

Patient → Game interaction → Labeled stimuli → WineBOX algorithm → Inferential profile → Clinical hypothesis

*Figure 1. Clinical systematization flow: from playful interaction to working hypothesis.*

## 2. Methodological Design

### 2.1. Algorithm Construction

#### *2.1.1. Stimulus control*

A database was constructed containing the complete catalog of WineBOX games (N=157). For each game, the following were recorded: name, stimuli (cognitive demand descriptors), and frequent clinical association (observational association with patterns reported in ASD/ADHD, using descriptive language observable only by the researchers). Initial coding was performed by a neuropsychologist and reviewed by a second researcher.

### *2.1.2. Assignment of weights by cognitive domain*

For each domain (Impulse Control, Sustained Attention, Planning, Cognitive Flexibility) certain keywords were defined. These are detailed in Table 1 and were determined by consensus from a panel of three experts (modified two-round Delphi procedure; Hsu & Sandford, 2007).

**Table 1** Weight matrix for the "Impulse Control" domain

| **Category** | **Keywords** | **Weight** | **Theoretical justification** |
|---|---|---|---|
| Quick response demand | fast, speed, action, immediate, reflexes, chaotic | +3 | Theoretically associated with inhibitory control demand (Logan et al., 1997) |
| Multitasking / quick switching | multitasking, split, simultaneous, fast switching | +2 | Frequent shifts in attentional focus (Monsell, 2003) |
| Planning demand | planning, strategy, organization, long term, sequencing | -2 | They reward reflection and anticipation (Owen, 1997) |
| Request for injunction | patience, turns, waiting, delay, inhibition | -2 | They need to suppress overbearing responses |
| Modulator: preference | "Yes" answer to "Would you play again?" | +1 | Amplifies the impact of the stimulus |
| Modulator: commitment | Difficulty ≥4 and feeling ≥4 in games with positive base weight | +2 | Enjoy the quick challenge |

*Note: Score per game = algebraic sum of keyword weights + modulators; range 0–10*

### *2.1.3. Calculation of scores*

For a patient with a certain number of games, we would have the following algorithmic inference: $P_{game} = \max(0, \sum(weight) + preference + commitment)$; $P_{total} = (\sum P_{game})/n$; normalization to 0–100: $P_{norm} = (P_{total}/10) \times 100$.

# 3. Sample

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## 3.1. Concordance study

**Participants:** Three judges: a PhD in Psychology, a clinical neuropsychologist, and an occupational therapist. None of them participated in the development of the algorithm; their evaluation was specifically in their capacity as judges.

**Stimuli:** 30 games were selected, through intentional sampling based on representativeness of domains and variability observed in the retrospective phase.

## 3.2. Clinical observation

**Data origin:** Between 2017 and 2025, user interaction records were collected from WineBOX in clinical and educational settings. The data were originally collected for clinical care purposes and subsequently anonymized for observational analysis. This procedure ensures the protection of personal data and compliance with privacy regulations (Argentine Law 25.326 on Personal Data Protection).

**Inclusion criteria:** (a) informed consent was submitted and had to be signed by parents or guardians; (b) the group included participants aged 7 (seven) to 17 (seventeen) years; (c) a minimum of 3 (three) completed sessions. A non-probability convenience sample was used (N=1,200; 68% Argentina, 12% Chile, 11% Mexico, 9% Spain).

**Age distribution:** Group 7–11 years: n=433 (36.1%); Group 12–17 years: n=767 (63.9%).

# 4. Instruments

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The WineBOX platform was used with its cognitive analysis module, which automatically records interactions. For the concordance study, judges rated the demand for "Impulse Control" and "Sustained Attention" for each game on a 1–5 Likert scale, based on a description and gameplay video.

# 5. Results

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## 5.1. Software and statistical analysis

All analyses were performed in R version 4.3.2 (R Core Team, 2023) using the packages *psych* v2.3.9 for reliability analysis, *lme4* v1.1-35 for exploratory mixed models, and *boot* v1.3-28 for calculating confidence intervals using bootstrap with 1000 replicates. The full scripts and source code are available in the supplementary material under the MIT license.

## 5.2. Verification of statistical data

Prior to the main analyses, the score distributions in the four cognitive domains were evaluated using the Shapiro-Wilk test. All domains met the assumption of normality (Impulse Control: $W=0.992$, $p=0.12$; Sustained Attention: $W=0.989$, $p=0.08$; Planning: $W=0.994$, p=0.21; Cognitive Flexibility: W=0.991, p=0.14), confirming hypothesis H4 and justifying the use of statistics.

## 5.3. Descriptive statistics of the sample

Normalized scores (0–100) were analyzed for the main domains. Table 2 summarizes the mean, standard deviation, 95% confidence interval, and observed range.

**Table 2** Descriptive statistics by cognitive domain

| **Domain** | **Average (DE)** | **95% CI (bootstrap)** | **Min–Max** |
|---|---|---|---|
| Impulse Control | 58.4 (21.3) | [57.2 – 59.6] | 12 – 98 |
| Sustained Attention | 52.7 (24.1) | [51.4 – 54.0] | 8 – 99 |
| Planning | 47.2 (19.8) | [46.1 – 48.3] | 5 – 95 |
| Cognitive Flexibility | 55.1 (22.6) | [53.9 – 56.3] | 10 – 97 |

*Note: Normalized scores from 0 to 100. Confidence intervals calculated using parametric bootstrap with 1000 replicates ( boot package in R).*

## 5.4. Inter-judge agreement

The analysis of the agreement of the data obtained between the three judges for the 30 stimuli showed the following coefficients:

**Table 3** Correlation Coefficients (ICC)

| **Domain** | **ICC(2,1)** | **95% CI** | **Classification (Koo & Li, 2016)** |
|---|---|---|---|
| Impulse Control | 0.79 | [0.71 – 0.86] | Good |
| Sustained Attention | 0.74 | [0.65 – 0.82] | Moderate to good |

Both domains exceeded the 0.70 threshold established in hypotheses H1 and H2, confirming good to excellent agreement. The complete judges' rating matrix is available in the supplementary material (OSF).

## 5.5. Correlations between domains (H3 contrast)

Pearson correlations were calculated between the scores of the different domains in the retrospective sample (N=1,200) to explore the interdependence of constructs. The correlation matrix is presented in Table 4.

**Table 4** Correlation matrix between cognitive domains

| | Impulse Control | Sustained Attention | Planning | Flexibility |
|---|---|---|---|---|
| Impulse Control | 1 | 0.52** | 0.31** | 0.44** |
| Sustained Attention | 0.52** | 1 | 0.38** | 0.49** |
| Planning | 0.31** | 0.38** | 1 | 0.41** |
| Flexibility | 0.44** | 0.49** | 0.41** | 1 |

*Note: ** p < 0.01. Pearson correlations with N=1,200.*

A moderate correlation can be observed between Impulse Control and Sustained Attention ($r=0.52$, $p < 0.01$), exceeding the threshold of $r \geq 0.40$ established in H3 and consistent with research and literature linking both constructs. However, Planning shows lower correlations with the other constructs, suggesting some independence as a state variable.

## 5.6. Observations by age groups

The 7- to 11-year-old group (n=433) showed greater variability in games with changes in the touch interface. Family reports suggested that prior exposure to video games could be a confounding variable. The 12- to 17-year-old group (n=767) showed the greatest variability in strategy/simulation games. The 17-year-old participants showed a preference for games with low visual fidelity but high cognitive load. These patterns guided the selection of the 30 stimuli for the concordance study.

# 6. Discussion

## 6.1. Interpretation of the main findings

The inter-rater reliability obtained (ICC $\geq 0.74$) confirmed hypotheses H1 and H2, providing preliminary evidence that the algorithm's stimulus classification aligns with expert clinical judgment. These values are comparable to those found in neuropsychological validation studies of observational instruments (Gioia et al., 2000). The slightly lower agreement for Sustained Attention could be due to the multidimensionality of the construct (Fortenbaugh et al., 2015).

However, the sample statistics show wide variability, which supports the instrument's sensitivity in detecting differences. Correlations between domains, particularly the relationship between Impulse Control and Sustained Attention ($r=0.52$), confirm hypothesis H3 and are consistent with theoretical models of executive functions (Miyake et al., 2000). The verification of normality within hypothesis H4 validates the use of parametric methods in future analyses.

## 6.2. Evidence

This study corresponds to a retrospective observational design, which, according to the standards of the Oxford Centre for Evidence-Based Medicine (OCEBM, 2011), is classified as **Level III evidence** . This classification places the findings as preliminary evidence useful for generating hypotheses, but which requires confirmation through higher-level prospective studies.

## 7. Clinical Implications

The results obtained have direct implications for clinical practice in child and adolescent neuropsychology:

- **Structuring the observation:** WineBOX offers a systematic framework for documenting behaviors during play interaction, reducing subjectivity and improving the traceability of observations.
- **Generation of working hypotheses:** The resulting inferential profile (normalized scores by domain) allows for the formulation of specific hypotheses about areas of difficulty or strengths, which can be explored in subsequent evaluations.
- **Longitudinal follow-up:** The quantitative and repeatable nature of the instrument facilitates the monitoring of changes over time, for example, in response to therapeutic interventions.
- **Ecological complement:** Unlike paper-and-pencil tests, WineBOX provides a playful context that approximates everyday situations, potentially improving the ecological validity of clinical inferences.

## 8. Limitations

**Non-diagnostic nature:** WineBOX is not a diagnostic tool; the profile is a working hypothesis. And it does not replace professional judgment.

**Construct validity pending:** Concurrent validation with standardized tests is required (Section 9 protocol).

**Uniform weighting:** Future versions will explore differential weights through automated semantic analysis.

**Reduced panel of judges:** n=3 is acceptable for exploratory study, but will be expanded in subsequent phases.

**Uncontrolled retrospective sample:** Data were originally collected for clinical purposes and subsequently anonymized for analysis.

**Confounding variable identified:** Prior exposure to video games will be controlled in future studies.

**Need for external replication and inter-institutional validation:** The results come from a single research team; replication by independent groups is required to confirm the robustness of the findings.

**Level of evidence:** As this is a retrospective observational study (Level III), the findings should be considered preliminary until confirmed by prospective designs.

# 9. Next Steps: Concurrent Validation and Reliability Protocol

## 9.1. Objectives

Evaluate concurrent validity (correlation with CPT-3, BRIEF-2, Tower of London).

## 9.2. Design and sample

Multifaceted study with groups with different pathologies: ADHD (n=50), ASD (n=50), and control (n=50), with an age range of 8 (eight) to 16 (sixteen) years, matched. The sample size to be able to detect correlations should be: $r \geq 0.40$ ($\alpha=0.05$, $\beta=0.20$).

## 9.3. Instruments

A variation of the WineBOX software (WineBOX Tool) will be used, oriented towards the rapid detection of patterns, complemented by the use of CPT-3, BRIEF-2, Tower of London, with the scale of prior exposure to video games, and the use of the demographic questionnaire.

## 9.4. Algorithm refinements

The use of latency in response, sequence analysis, and the improved emotion regulation module will be incorporated.

## 10. Conclusion

WineBOX systematizes the clinical observation of digital gaming environments based on Cognitive Load Theory. The results confirm the four hypotheses: inter-rater agreement ≥ 0.70 (Hypotheses H1 and H2), moderate correlation between domains (Hypothesis H3), and normality of distributions (Hypothesis H4). Descriptive statistics from the 1,200 cases support the instrument's content validity and sensitivity. The system is not diagnostic; it only presents a spatial representation of domains, therefore further validation is required.

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## Supplementary Material (open access)

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Available at https://github.com/iflex/Freedows8Bit

- Appendix A: Complete Judge Rating Matrix (30 stimuli, 3 judges) – CSV.
- Appendix B: Statistical analysis script in R for ICC, correlations and bootstrap.
- Appendix C: Informed consent questionnaire approved by CEI.
- Appendix D: Complete catalog of games (N=157) with stimulus labeling – JSON.
- Appendix E: Standardized administration protocol.
- Appendix F: Anonymized database (N=1,200) for replication.

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